Alg2 Notes 8-4: Simplifying, Multiplying, and Dividing Rational Expressions

~~*Note! HW says "State any restrictions on the variables". You do not need to do this, yet*~~.

I. Simplifying: Anything divided by itself = 1

Allows us to Cancel

$\frac{12}{16} = \frac{(3)\cancel{(4)}}{(4)\cancel{(4)}}$ $\frac{5}{5} = 1$ $\frac{(x+1)}{(x+1)} = 1$ $\frac{\cancel{(x+2)}(x-3)}{(x+4)\cancel{(x+2)}} = \frac{x-3}{x+4}$

You can only cancel out quantities if you have a product (result of multiplication) or value can go into all terms

CAN'T : $\frac{x+2}{2}$ No! DON'T.

CAN: $\frac{2x}{2} = x$

Can: $\frac{2x+6}{2}$ $= \frac{\cancel{2}(x+3)}{\cancel{2}}$ $= x+3$

What do you think $\frac{3x+6}{2}$? Nothing Cancels

Ex1: Simplify: $\frac{x^2+7x+10}{x^2-3x-10} =$

FACTOR then Simplify

$= \frac{(x+5)\cancel{(x+2)}}{(x-5)\cancel{(x+2)}}$

$= \boxed{\frac{x+5}{x-5}}$

You! Simplify: $\frac{x^2+2x-8}{x^2-5x+6}$

$\frac{x^m}{x^n} = x^{m-n}$

$x^{-2} = \frac{1}{x^2}$

Ex2: Simplify: $\frac{24x^3y^2}{-6x^2y^3} =$

$\frac{(4)\cancel{(6)}x^3y^2}{(-1)\cancel{(6)}x^2y^3} = \frac{-4x}{y}$

You! Simplify: $\frac{2x^4y^3}{18x^2y^0}$

$\frac{x^2y^3}{9}$

II Multiplying: Factor & Cancel!

smush together

Ex3. $\frac{\cancel{(x+2)}(x+1)}{(x+3)} \cdot \frac{(x-4)}{\cancel{(x+2)}}$

one big bar!

$= \frac{(x+1)(x-4)}{(x+3)}$

Ex4: $\frac{(x-3)}{\cancel{(x+2)}} \cdot \frac{\cancel{(x+1)}\cancel{(x+2)}}{\cancel{(x+1)}}$

$= x-3$

Ex5. $\frac{x^2+x-6}{x-5}\cdot\frac{x^2-25}{x^2+4x+3}$

You! Simplify: $\cancel{\frac{x^2-2x-24}{x^2+7x+12}\cdot\frac{x^2-1}{(x-6)}}$

Factor before putting the big bar

$$\frac{\cancel{(x+3)}(x-2)}{\cancel{(x-5)}}\cdot\frac{(x+5)\cancel{(x-5)}}{\cancel{(x+3)}(x+1)}$$

$$=\frac{(x-2)(x+5)}{x+1}$$

III Dividing: Flip 2nd fraction, Factor & Cancel!

mult

Ex5. $\frac{x^2+5x+4}{x^2+x-6}\div\frac{x^2-1}{2x^2-6x}$

$$\frac{(x+4)(x+1)}{(x+3)(x-2)}\cdot\frac{(2x^2-6x)}{(x^2-1)}$$

$$\frac{(x+4)\cancel{(x+1)}(2x)(x-3)}{(x+3)(x-2)\cancel{(x+1)}(x-1)}$$

$$=\frac{2x(x+4)(x-3)}{(x+3)(x-2)(x-1)}$$

You! Simplify: $\frac{3x-12}{2x+4}\div\frac{6x-24}{8+4x}$

$$\frac{\cancel{3}\cancel{(x-4)}}{\cancel{2}\cancel{(x+2)}}\cdot\frac{\cancel{4}(2+x)}{\cancel{6}\cancel{(x-4)}}$$

$$=1$$

Ex6. $\frac{2-x}{x^2+2x+1}\div\frac{x^2+3x-10}{x^2-1}$

$$\frac{(2-x)}{(x+1)\cancel{(x+1)}}\cdot\frac{\cancel{(x+1)}(x-1)}{(x+5)(x-2)}$$

$$\frac{-1\cancel{(x-2)}(x-1)}{(x+1)(x+5)\cancel{(x-2)}}$$

$$=\frac{(-x+1)}{(x+1)(x+5)}$$

can't cancel

You! Simplify: $\cancel{\frac{2x-8}{16-x^2}\div\frac{x^2+5x+4}{x^2+8x+16}}$

The negative one trick!

$(-1)(-1)(2-x)$ Distribute

+1 so legal

$(-1)(-2+x)$ swap

$(-1)(x-2)$

$2-x=-1(x-2)$

~~Same~~ 2 terms with subtraction can turn into